Dauphine UNIVERSITÉ | PARIS SCIENCES & LETTRES

REVIEW TITLE GENERATION: TEXT SUMMARIZATION WITH ENCODER-DECODER MODELS

NLP course project presented by

Omar Belhaj

SELECTED TASK

Summarizing a review in a few words

- Task: extreme abstractive summarization
- Input: an Amazon food review
- Output: a title of about 3 words, generated word by word
- Goal: keep both the subject and the opinion

Example :

Review: This tea is not good at all. It tastes like nothing and I would not recommend it.
Summary: not good

Why it matters: faster reading, suggested titles, frequent complaints spotted at a glance.

SELECTED TASK

Review

REVIEW

I was really looking forward to trying this yogurt after seeing so many positive advertisements, but it completely failed to live up to the hype. First of all, the texture is incredibly strange. Instead of being thick, smooth, and creamy like a quality yogurt should be, it has a weirdly gelatinous, almost watery consistency. The moment you open the lid, there is a thick layer of separated liquid on top, and even after giving it a vigorous stir, it still feels chalky and thin in your mouth.

Result

GENERATED TITLE

not so good

THE DATASET

Amazon Fine Food Reviews

Stanford Network Analysis Project and 1 collaborator · Updated 10 years ago

2469 | Code | Download

Amazon Fine Food Reviews

Analyze ~500,000 food reviews from Amazon



Data Card | Code (1214) | Discussion (17) | Suggestions (0)

About Dataset

Context

This dataset consists of reviews of fine foods from amazon. The data span a period of more than 10 years, including all ~500,000 reviews up to October 2012. Reviews include product and user information, ratings, and a plain text review. It also includes reviews from all other Amazon categories.

Contents

- Reviews.csv: Pulled from the corresponding SQLite table named Reviews in database.sqlite
- database.sqlite: Contains the table 'Reviews'

Usability
7.94

License
CC0: Public Domain

Expected update frequency
Not specified

Tags
Business | Internet

THE DATASET

Amazon Fine Food Reviews

Step	Reviews
Raw dataset	568,454
After removing duplicates	393,552
After length filters	317,758
Sampled for the first models	60,000

THE DATASET

Amazon Fine Food Reviews

Split by product: 47,762 train · 6,137 validation · 6,101 test

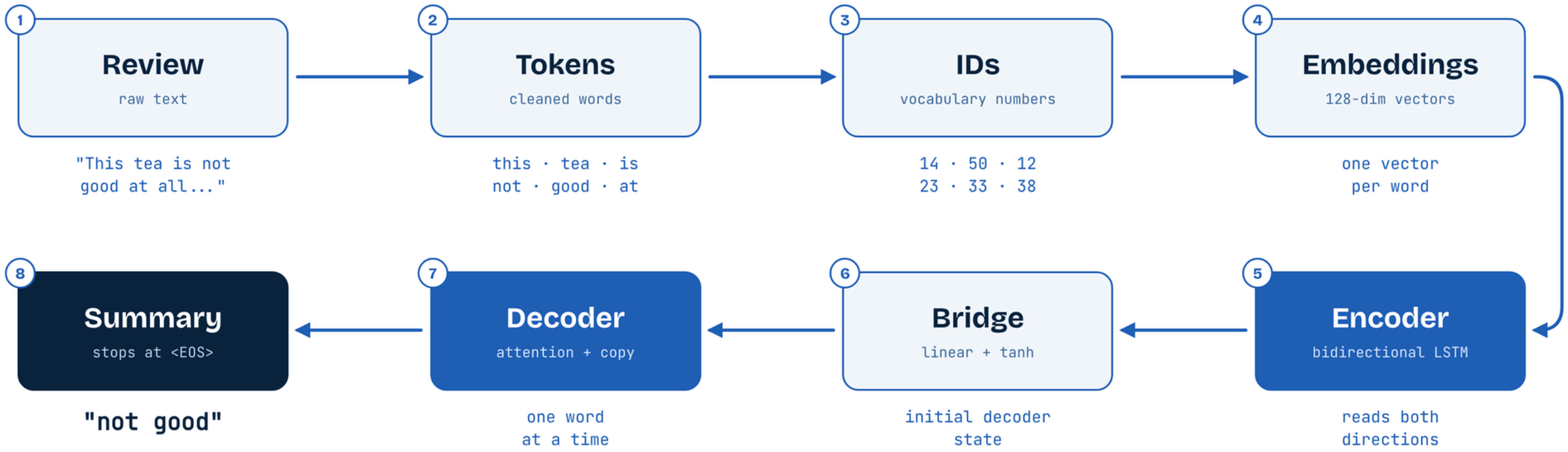
Vocabularies (train only): 16,429 review words · 4,342 title words

Special tokens: <PAD> <UNK> <SOS> <EOS>

Challenges: 78.9% positive reviews, many generic titles such as « great product »

THE ARCHITECTURE

From review to summary : the encoder-decoder pipeline



THE ARCHITECTURE

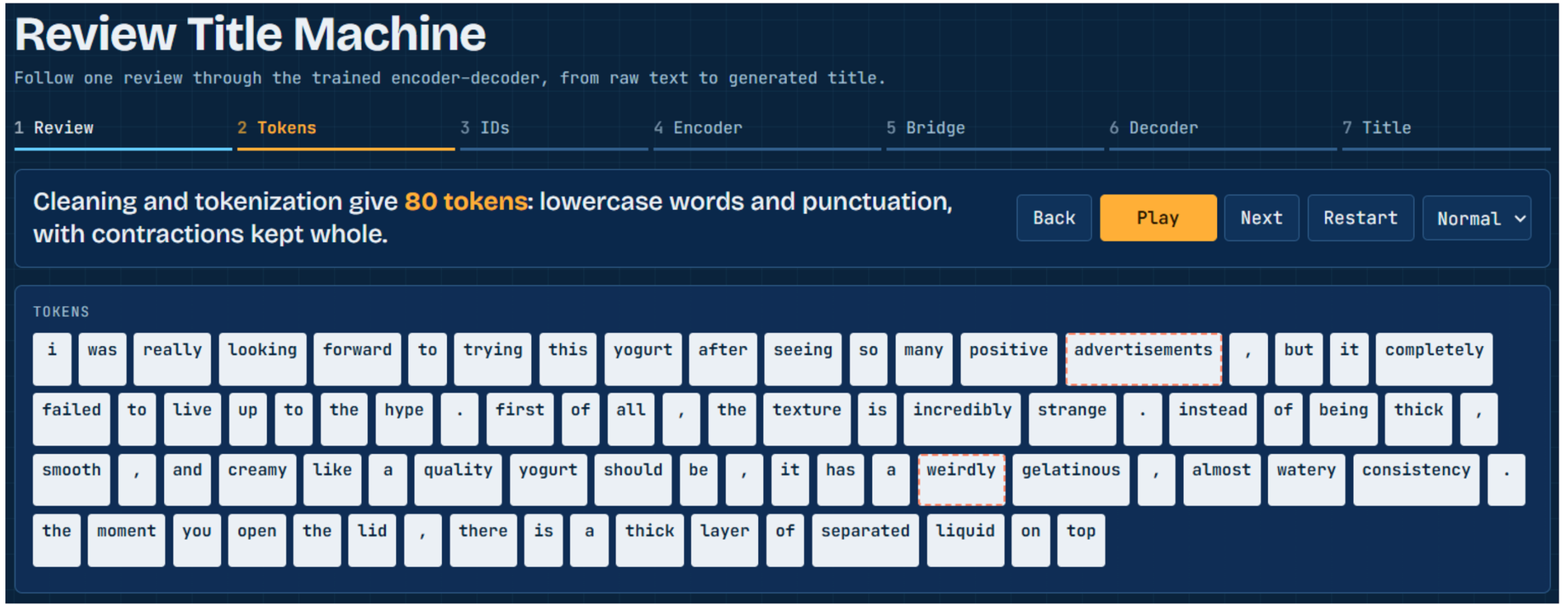
Let’s explain the architecture with this review example :

REVIEW

I was really looking forward to trying this yogurt after seeing so many positive advertisements, but it completely failed to live up to the hype. First of all, the texture is incredibly strange. Instead of being thick, smooth, and creamy like a quality yogurt should be, it has a weirdly gelatinous, almost watery consistency. The moment you open the lid, there is a thick layer of separated liquid on top, and even after giving it a vigorous stir, it still feels chalky and thin in your mouth.

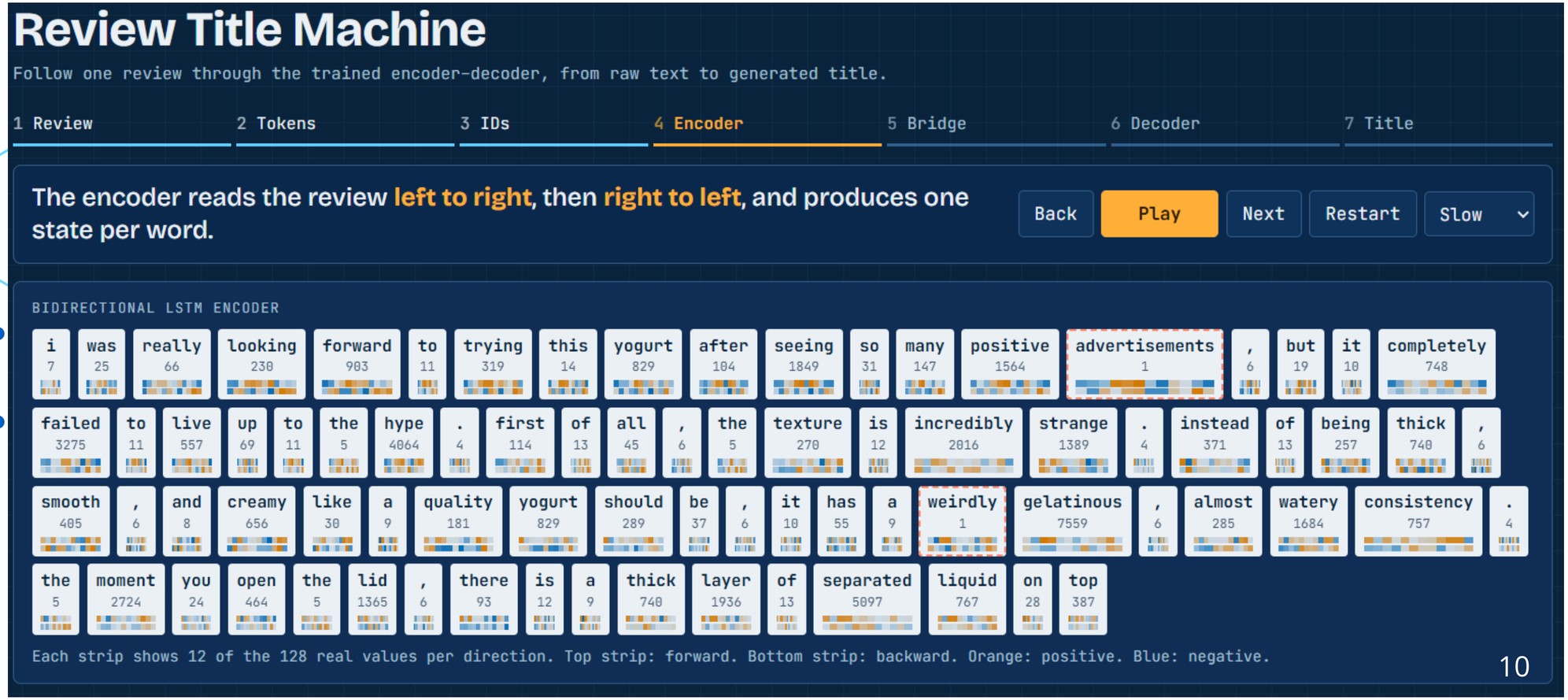
THE ARCHITECTURE

The review is cleaned and split into **tokens**, each token is replaced by its number in the vocabulary, because the model only works with numbers, and unknown words become <UNK>.



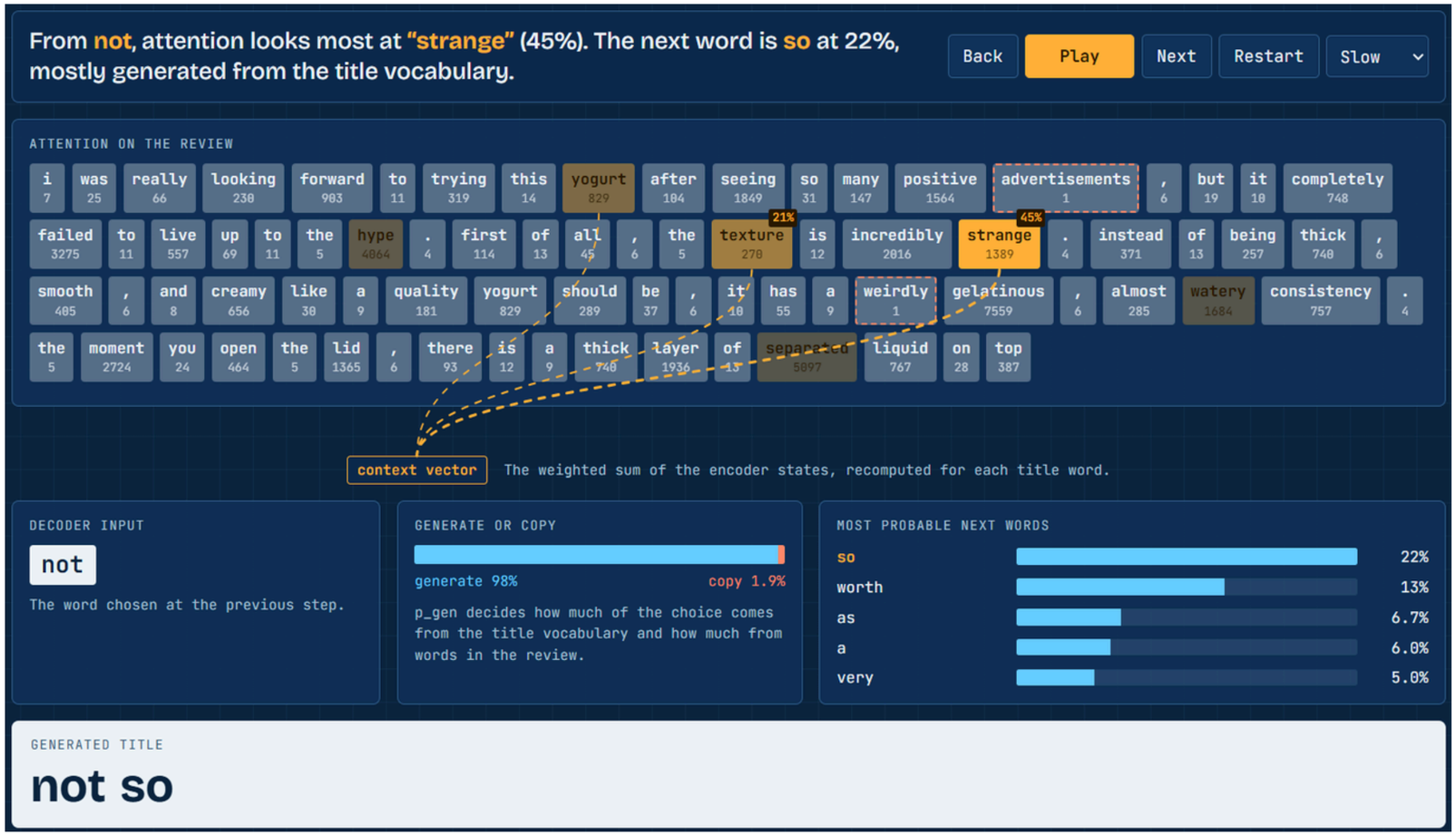
THE ARCHITECTURE

Encoder : a bidirectional LSTM reads the review in both directions



THE ARCHITECTURE

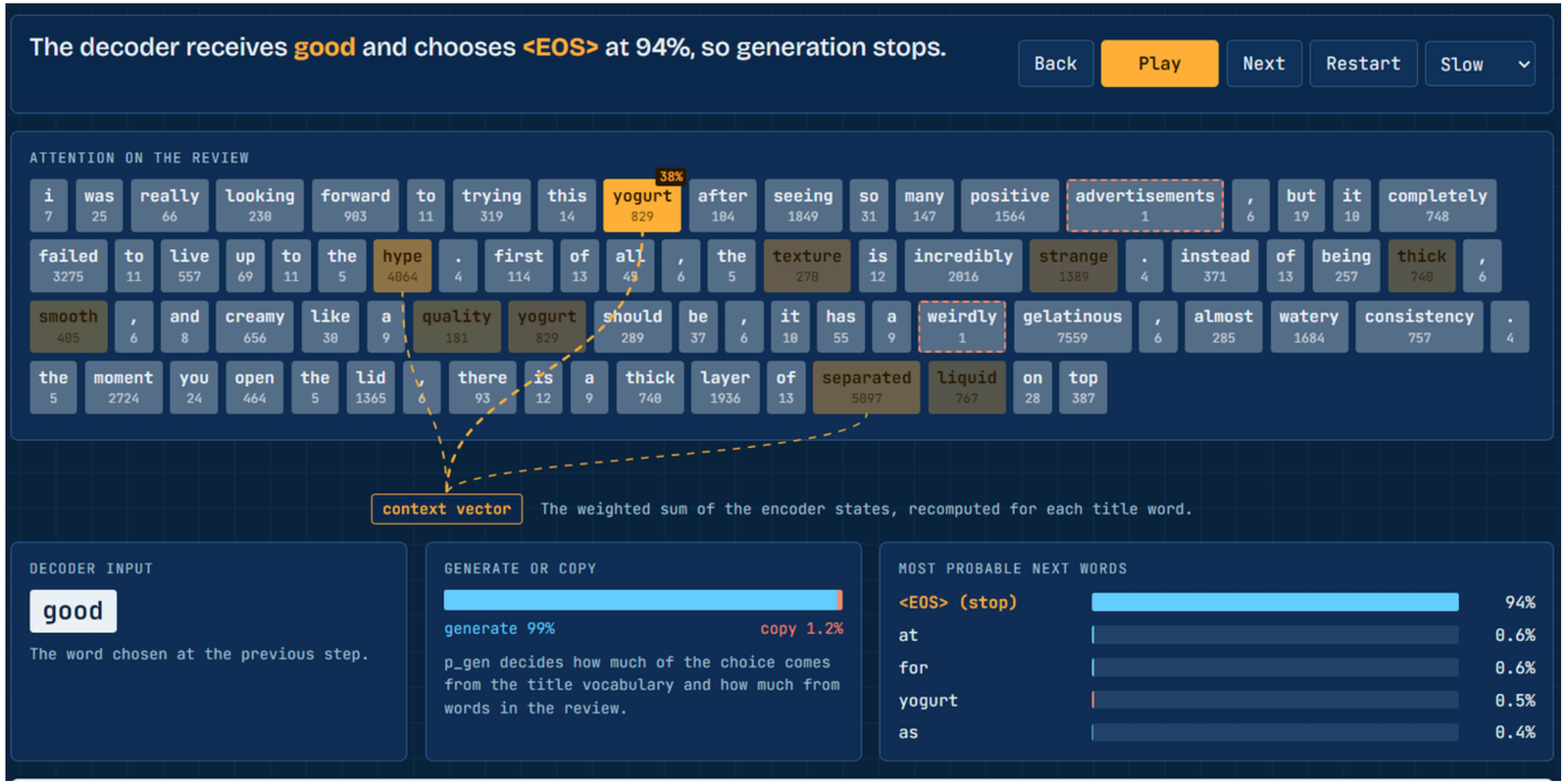
Bridge: turns the encoder's final state into the decoder's starting state



THE ARCHITECTURE

Decoder: an LSTM that writes the title one word at a time, from <SOS> to <EOS>

Attention: at each step, it focuses on the most useful review words



THE ARCHITECTURE

Output :

GENERATED TITLE

not so good

Copy mechanism: it can reuse a word from the review, such as a product name

Example :

Very delicious lasagna ! From the very first bite, I knew this wasn't just a regular meal, it was a true culinary masterpiece. The presentation alone was stunning, served piping hot with the cheese on top perfectly melted and beautifully golden-brown around the edges. When I cut into it, the layers stayed perfectly intact, showcasing the incredible craftsmanship that went into building this dish.

Result :

GENERATED TITLE

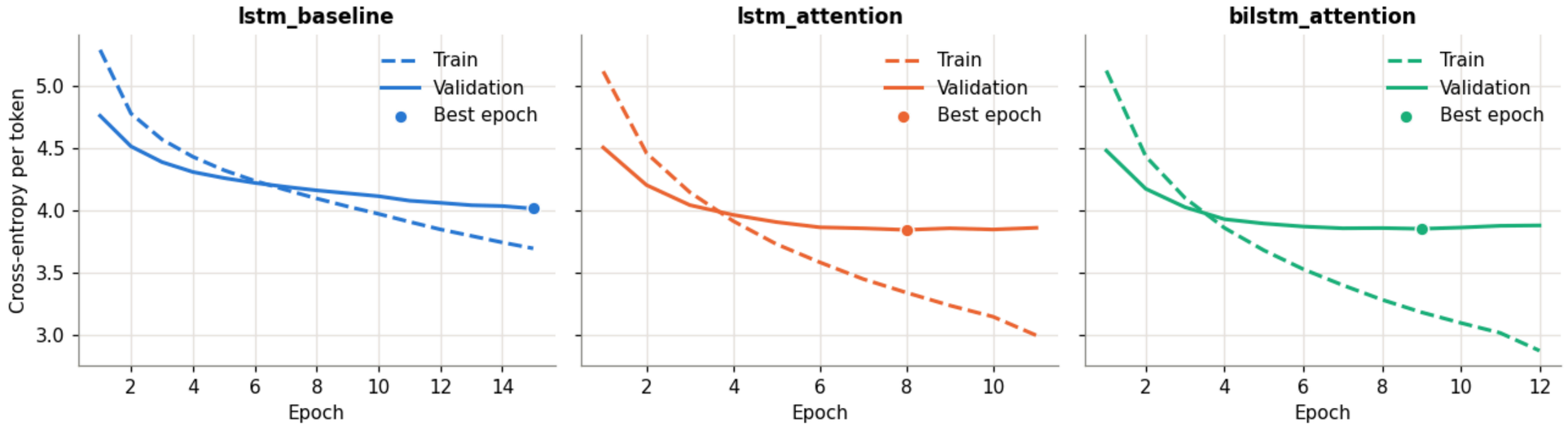
great lasagna

THE TRAINING PROCEDURE

Same settings for every model

- **Loss:** token-level cross-entropy
- **Optimizer:** Adam, learning rate 0.001
- **Batch size:** 128, up to 15 epochs, early stopping
- **Teacher forcing:** the decoder receives the true previous word during training
- **Generation:** greedy decoding (beam search also tested)
- **Hardware:** T4 GPU on Colab

THE TRAINING PROCEDURE



Attention models reach a lower validation loss, but start to overfit after about epoch 8.

EXPERIMENTS

Model	Change
LSTM baseline	No attention
LSTM + attention	Adds attention
BiLSTM + attention	Bidirectional encoder
Final model	+ copy mechanism + balanced sampling + 150,000 reviews

RESULTS ON TEST SET

Model	ROUGE-1	Distinct titles	Sentiment accuracy
First 4 words (no learning)	0.094	85.9%	0.635
LSTM baseline	0.105	0.5%	0.822
LSTM + attention	0.153	16.7%	0.753
BiLSTM + attention	0.159	23.1%	0.754
Final model	0.183	41.9%	0.830

GENERATED EXAMPLES

Before and after the improvements

Review (shortened)	Before	Final model
Dog food in a broken bag, my dog refused to eat it	dog food	dog hated it ✅
This tea is not good at all	not as good as the tea	not good ✅
Flavor is great, but the price is far too high	good but not great	good flavor but too high ✅
Expected chips to be terrible, but really tasty	not bad	terrible taste ❌

LIMITATIONS

Limitations and possible improvements

Limitation	Possible improvement
Misses contrast (« terrible taste » for a tasty product)	More data, Transformer encoder
Generic summaries (« great product »)	All eligible reviews, pretrained embeddings
25.5% of negative reviews get a positive summary	Tune the sampling weights
ROUGE misses meaning	Larger manual evaluation

CONCLUSION

Main findings

- A from-scratch encoder–decoder can summarize a review in a few words
- Attention and more data bring the main gains: ROUGE-1 from 0.105 to 0.183
- Remaining errors come mostly from sentence structure and data imbalance

THANK YOU !

Github Link :
https://github.com/omar-belhaj/nlp-review-title-generation